

Task on Functions in Python

1. Write a function `greet(name)` that takes a person's name as input and returns a greeting message.
Example:
Input: `greet("Alice")`
Output: `"Hello, Alice!"`
2. Write a function `is_even(number)` that checks if a number is even. The function should return `True` for even numbers and `False` for odd numbers.
Example:
Input: `is_even(4)`
Output: `True`
3. Write a function `find_max(a, b)` that takes two numbers as input and returns the larger of the two.
Example:
Input: `find_max(10, 20)`
Output: `20`
4. Write a function `calculator(a, b, operation)` that performs basic arithmetic operations (add, subtract, multiply, divide) based on the `operation` string provided.
Example:
Input: `calculator(5, 3, 'add')`
Output: `8`
5. Write a function `factorial(n)` that computes the factorial of a given non-negative integer `n`.
Example:
Input: `factorial(5)`
Output: `120`
6. Write a function `is_prime(n)` that checks if a number is a prime number. Return `True` if it is, otherwise return `False`.
Example:
Input: `is_prime(11)`
Output: `True`
7. Write a function `count_vowels(s)` that counts the number of vowels in a given string.
Example:
Input: `count_vowels("hello")`
Output: `2`
8. Write a function `fibonacci(n)` that generates the first `n` numbers in the Fibonacci sequence.
Example:
Input: `fibonacci(5)`
Output: `[0, 1, 1, 2, 3]`
9. Write a function `is_palindrome(s)` that checks if a string is a palindrome (reads the same forwards and backwards).
Example:

Input: `is_palindrome("madam")`

Output: `True`

10. Write a function `bubble_sort(arr)` that sorts a list of numbers in ascending order using the bubble sort algorithm.

Example:

Input: `bubble_sort([4, 2, 7, 1])`

Output: `[1, 2, 4, 7]`

11. Write a function `are_anagrams(s1, s2)` that checks if two strings are anagrams of each other.

Example:

Input: `are_anagrams("listen", "silent")`

Output: `True`

12. Write a function `find_duplicates(lst)` that identifies all the duplicate elements in a list and returns them in a new list.

Example:

Input: `find_duplicates([1, 2, 3, 2, 4, 5, 1])`

Output: `[1, 2]`

13. Write a function `convert_temperature(value, unit)` that converts a temperature value from Celsius to Fahrenheit or vice versa based on the unit provided.

Example:

Input: `convert_temperature(0, 'C')`

Output: `32`

14. Write a function `calculate_bmi(weight, height)` that calculates the Body Mass Index (BMI) based on weight in kilograms and height in meters.

Example:

Input: `calculate_bmi(70, 1.75)`

Output: `22.86`

15. Write a function `is_valid_email(email)` that checks if a given email address is valid based on its basic format (e.g., it should contain @ and a domain).

Example:

Input: `is_valid_email("example@gmail.com")`

Output: `True`

16. Write a function `calculate_distance(x1, y1, x2, y2)` that calculates the Euclidean distance between two points in a 2D plane.

Example:

Input: `calculate_distance(1, 2, 4, 6)`

Output: `5.`

17. Write a function `square(num)` that takes a number as input and returns its square.

Example:

Input: `square(5)`

Output: `25`

18. Write a function `reverse_string(s)` that takes a string as input and returns the string reversed.

Example:

Input: `reverse_string("hello")`

Output: `"olleh"`

19. Write a function `is_positive(num)` that checks if a number is positive, negative, or zero and returns a corresponding message.
Example:
Input: `is_positive(10)`
Output: "Positive"
20. Write a function `sum_of_list(lst)` that takes a list of numbers and returns the sum of its elements.
Example:
Input: `sum_of_list([1, 2, 3, 4])`
Output: 10
21. Write a function `unique_elements(lst)` that takes a list of numbers and returns a new list with only the unique elements.
Example:
Input: `unique_elements([1, 2, 2, 3, 4, 4])`
Output: `[1, 2, 3, 4]`
22. Write a function `capitalize_words(sentence)` that takes a sentence and capitalizes the first letter of each word.
Example:
Input: `capitalize_words("hello world")`
Output: "Hello World"
23. Write a function `count_occurrences(lst, item)` that counts how many times a specific element appears in a list.
Example:
Input: `count_occurrences([1, 2, 3, 2, 4, 2], 2)`
Output: 3
24. Write a function `armstrong_number(n)` that checks if a number is an Armstrong number.
(An Armstrong number is a number equal to the sum of its own digits raised to the power of the number of digits.)
Example:
Input: `armstrong_number(153)`
Output: True
25. Write a function `merge_sorted_lists(lst1, lst2)` that merges two sorted lists into a single sorted list.
Example:
Input: `merge_sorted_lists([1, 3, 5], [2, 4, 6])`
Output: `[1, 2, 3, 4, 5, 6]`
26. Write a function `longest_word(sentence)` that returns the longest word in a given sentence.
Example:
Input: `longest_word("The quick brown fox jumps over the lazy dog")`
Output: "jumps"
27. Write a function `generate_primes(n)` that generates a list of prime numbers less than or equal to `n`.
Example:

Input: `generate_primes(10)`

Output: `[2, 3, 5, 7]`

28. Write a function `remove_punctuation(s)` that removes all punctuation from a given string.

Example:

Input: `remove_punctuation("Hello, world!")`

Output: `"Hello world"`

29. Write a function `password_strength(password)` that checks the strength of a password based on certain criteria (e.g., length, presence of uppercase letters, numbers, and symbols).

Example:

Input: `password_strength("Pass123!")`

Output: `"Strong"`

30. Write a function `shopping_list_total(items, prices, quantities)` that calculates the total cost of items in a shopping list based on their prices and quantities.

Example:

Input: `shopping_list_total(["apple", "banana"], [1.5, 0.5], [3, 4])`

Output: `7.5`

31. Write a function `student_report(name, marks)` that accepts a student's name and a list of marks, calculates the average, and assigns a grade.

Example:

Input: `student_report("John", [80, 75, 90, 85])`

Output: `"John's Average: 82.5, Grade: B"`

32. Write a function `generate_password(length)` that generates a random password containing uppercase, lowercase, digits, and special characters.

Example:

Input: `generate_password(8)`

Output: `"A$g8x!Y2"`

33. Write a function `shopping_cart(items)` that takes a list of item prices and returns the total cost with a 10% discount applied if total > 1000.

Example:

Input: `shopping_cart([250, 400, 600])`

Output: `1125.0`

34. Write a function `group_by_even_odd(numbers)` that separates a list into even and odd numbers.

Example:

Input: `group_by_even_odd([1, 2, 3, 4, 5])`

Output: `([2, 4], [1, 3, 5])`

35. Write a function `email_filter(emails)` that returns a list of valid Gmail addresses from a given list of emails.

Example:

Input: `email_filter(["a@gmail.com", "b@yahoo.com", "c@gmail.com"])`

Output: `["a@gmail.com", "c@gmail.com"]`

36. Write a function `count_words(sentence)` that counts the number of each word in the sentence and returns a dictionary.

Example:

- Input: `count_words("hello world hello")`
Output: `{"hello": 2, "world": 1}`
37. Write a function `is_strong_password(password)` that checks if a password is at least 8 characters long and includes letters, numbers, and a special character.
Example:
Input: `is_strong_password("Pass123!")`
Output: `True`
38. Write a function `convert_to_dict(keys, values)` that takes two lists and returns a dictionary mapping keys to values.
Example:
Input: `convert_to_dict(["a", "b"], [1, 2])`
Output: `{"a": 1, "b": 2}`
39. Write a function `simulate_atm(balance, operations)` that takes an initial balance and a list of operations (deposit/withdraw), then returns the final balance.
Example:
Input: `simulate_atm(1000, [("deposit", 500), ("withdraw", 300)])`
Output: `1200`
40. Write a function `is_valid_pin(pin)` that checks if the pin is a 4-digit or 6-digit number.
Example:
Input: `is_valid_pin("1234")`
Output: `True`
41. Write a function `temperature_stats(temps)` that calculates the highest, lowest, and average temperature from a list.
Example:
Input: `temperature_stats([23, 30, 19, 28])`
Output: `(30, 19, 25.0)`
42. Write a function `grade_distribution(marks)` that returns the number of students in each grade category (A, B, C, D, Fail).
Example:
Input: `grade_distribution([92, 81, 75, 66, 58])`
Output: `{"A": 1, "B": 1, "C": 1, "D": 1, "Fail": 1}`
43. Write a function `character_frequency(s)` that counts how often each character appears in a string.
Example:
Input: `character_frequency("apple")`
Output: `{"a": 1, "p": 2, "l": 1, "e": 1}`
44. Write a function `calculate_interest(principal, rate, years, compound)` that returns the total interest based on whether it's simple or compound interest.
Example:
Input: `calculate_interest(1000, 5, 2, 'compound')`
Output: `102.5`
45. Write a function `student_summary(data)` that takes a list of dictionaries containing name and score and returns the name of the top student.
Example:
Input: `[{"name": "A", "score": 85}, {"name": "B", "score": 92}]`
Output: `"B"`

46. Write a function `get_pricing_summary(cart)` that takes a dictionary of items with prices and quantities and returns the total price.
Example:
Input: `{"apple": (30, 2), "banana": (10, 5)}`
Output: 110
47. Write a function `validate_username(username)` that checks if the username is alphanumeric and between 5–12 characters.
Example:
Input: `validate_username("user_01")`
Output: False
48. Write a function `average_nested_list(nested_list)` that calculates the average of all numbers in a list of lists.
Example:
Input: `average_nested_list([[1, 2], [3, 4, 5]])`
Output: 3.0
49. Write a function `sort_students_by_grade(students)` that sorts a list of tuples (name, grade) in descending order of grades.
Example:
Input: `[("Alice", 85), ("Bob", 92)]`
Output: `[("Bob", 92), ("Alice", 85)]`
50. Write a function `word_count_summary(paragraph)` that returns the number of words, characters, and unique words.
Example:
Input: `"hello world hello"`
Output: `{"words": 3, "characters": 17, "unique_words": 2}`